A deep learning based neuro-fuzzy approach for solving classification problems

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***Abstract*— Techniques involved artificial intelligence and machine learning offers various classification methods in order to deal with daily life problems. Among these methods, Adaptive Neuro-Fuzzy Inference Systems (ANFIS) and Deep Neural Network (DNN) are the most commonly used classifiers. Since ANFIS is not suitable for high-dimensional data, therefore DNN was introduced to overcome this problem faced by conventional methods. However, due to the optimization of millions of parameters in their deep architecture, the decision made by DNN faced the criticism of being non-transparent. To overcome this problem, recently, various researchers are coming up with the idea of using fuzzy logic with DNN. Therefore, this study also proposed a Deep Neuro-Fuzzy Classifier (DNFC) with a cooperative based structure for solving classification problems, particularly. The performance of the proposed DNFC was evaluated with ANFIS and DNN classifier, where overall results show that the performance of ANFIS classifier decreased when input size increased. While the performance of the proposed model demonstrated nearly similar or slightly higher accuracy as compared to DNN.**

Keywords—machine learning, adaptive neuro-fuzzy inference systems, deep learning, deep neural network, deep neuro-fuzzy systems.

I. INTRODUCTION

Techniques under Artificial Intelligence (AI) and Machine Learning (ML) indicates a computer to execute the tasks similar to human cognition. Generally, AI methods are comprised of Expert Systems (ES), Artificial Neural Networks (ANN), Fuzzy Logic (FL), and hybrid approaches such as neuro-fuzzy systems, genetic programming neural networks and many more [1]. Among these methods, Adaptive Neuro-Fuzzy Inference Systems (ANFIS) and Deep Neural Network (DNN) are two well-known approaches to solve various real-world problems such as speech recognition, biometric identification, document classification, etc.

The ANFIS model is one of the modern hybrid systems of AI which combines the advantages of neural network and fuzzy logic in a single network. This model allows the building of IF-THEN rules and membership functions from the given data, as well as; it has learning ability to adapt new values by itself [2]. That means the ANFIS model maintains the transparency of the fuzzy system and the learning capacity that is inherited from neural networks [3]. ANFIS uses the hybrid of backpropagation and least square estimator based learning algorithm for the fast convergence. Since its existence, ANFIS has demonstrated great performance in tasks including education, medical, economic system, traffic control, forecasting and various others [4]. Despite its implementation in several applications, the ANFIS model faces computational complexity since it needs to translate former information into the network topology, which makes it resistant to the number of input variables [5].

This problem challenged research communities since the digital data was growing exponentially, and it became essential to develop the systems that can handle high dimensional data. Therefore, DNN was introduced in 2006 and based on its tremendous performance, the application of DNN increased rapidly due to its higher level of abstraction, which was challenging to be managed using conventional techniques [6]. DNN rises swiftly as a promising approach in research due to the handling of images, sound, and text data [7]. Before the arrival of DNN, the majority of ML systems have to go through manual feature extraction and classification. Whereas DNN can learn the features that optimally represent the given training data [8]. DNN has demonstrated outstanding results in various applications, including speech recognition, computer vision, and natural language processing [9]. However, regardless of their success, DNN models are claimed as “black-box” because the model trained using DNN are not easily understandable due to their deep architecture where millions of parameters are calculated and optimized by the network making predictions non-transparent and difficult to trace by humans [10].

Nevertheless, since fuzzy rules make feature extraction more human interpretable under uncertainty, various researchers incorporated the concepts of fuzzy logic into DNN as deep neuro-fuzzy systems. The combination brings together the advantages of both fuzzy logic and DNNs in extracting useful high-level features from ambiguous data. This model has the capability of ANFIS for handling uncertainty, as well as the advantages of deep learning [11]. The concept of combining fuzzy inference systems with deep neural networks has been implemented in various application areas such as; healthcare, traffic flow prediction and incident detection, data, text, video and image classification and recognition [12].

II. RESEARCH BACKGROUND

The machine learns from raw data during classification and then utilizes this learning to classify new observations. Various classifiers have been successfully implemented in the literature to solve several classification problems, such as Naive Bayes (NB), Support Vector Machine (SVM), K-Nearest Neighbor (KNN). Since the focus of this study is on deep neuro-fuzzy systems, which is a combination of ANFIS and DNN classifiers, therefore, this section will be explaining the architecture and concept behind these classifiers along with deep neuro-fuzzy classifier.

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A. Adaptive Neuro-Fuzzy Inference Systems (ANFIS)

The majority of the real-world classification problems have nonlinear behavior; therefore, the Adaptive Neuro-Fuzzy Inference Systems (ANFIS) have been widely used in various problems with nonlinearity to produce better classification results [13]. ANFIS was introduced by Jang in 1993 with the integration of FL and ANN [14]. According to Zadeh [15], instead of using binary logic that can be either True or False (0 or 1), FL gives a promising solution to real-world challenges by implementing the boolean logic; a notion of values that can be partially true and partially false using membership degree between 0 and 1. ANFIS uses fuzzy logic with the concept of fuzzy IF-THEN rules that helps to well-define the vagueness of human thinking [16]. While ANN benefits from its learning experience due to the handling of information in a similar way to the human brain [17]**.** Hence, the ANFIS model can incorporate human knowledge along with self-learning competencies that can potentially approximate every situation**.**

The ANFIS model consists of a five-layer architecture as; (i) input layer, (ii) input MF layer, (iii) fuzzy rules layer, (iv) output MF layer, (v) defuzzification layer. The nodes of each layer are connected to another layer by directed links. Therefore, to produce the output for a single node, each node performs a particular function on its incoming signals. Hence it is often known as a multilayer feed-forward network. The main objective of the ANFIS is to determine the optimum values of the equivalent fuzzy inference system parameters by applying a learning algorithm. To achieve minimum error, the ANFIS learning algorithm updates trainable parameters such as; membership function and consequent parameters using two-pass hybrid learning algorithms with Gradient Descent (GD) and Least Square Estimator (LSE). During the forward pass, LSE tunes consequent parameters, while in the backward pass, GD updates membership function parameters to further approximate the model with better accuracy [18].

B. Deep Neural Network (DNN)

A DNN network is assembled in a sequence of multiple layers that carries a collection of neurons, where the input activations from the previous layer are given to the neurons of the next layers for basic calculation. The neurons of the network jointly implement a complex nonlinear mapping from the input to the output. This mapping is learned from the data by adapting the weights of each neuron using a technique called error backpropagation [19].

Deep Neural Network (DNN) consists of single input, single output and multiple hidden layers. It allows different levels of abstraction to change the number, size and structure of each layer and enables high-level features to be extracted from low-level features to create a hierarchical representation [20]. Generally, a single layer carries several nodes, where each node is linked by a predetermined set of weights from the former layers. The selection of weights is a critical part and is carried out during the training period. By allocating values to the inputs and feed-forwarding them through the network to get the final output, the values of each layer can be determined from previous layer nodes. Whereas, the value of each hidden node in the network is measured by calculating a linear combination of node values from the previous layers and then using a nonlinear activation function. Once an activation function is applied to a node, the value of that node is computed as the maximum of the linear combination of the node from the previous layer and 0 [21].

C. Deep Neuro-Fuzzy Systems (DNFS)

Deep learning (DL) methods such as Deep Neural Networks (DNN) are an emerging and dominant paradigm that allows large-scale task-driven feature learning from big data [22]. This approach has made significant success in various tasks, including image and video recognition, audio processing, text analysis, economics and financial predictions and robotics [23], [24]. Though these approaches are powerful, they do not explain how the feedback is produced because of their black-box nature, which ultimately limits the applications of deep learning methods. It is not possible to verify the decision made by the model without understating of how they reach to certain outcome [25]. Nevertheless, this shortcoming of DNN opened new doors for researchers to explore the various effective ways to overcome the "black-box" issue. One of the promising solutions found by researchers was to incorporate the concepts of the explainable rule-based structure called fuzzy inference with DNN as a Deep Neuro-Fuzzy System (DNFS). It is seen that using fuzzy theory, along with deep learning can improve the performance of the models where data are noisy, heterogeneous, incomplete, or vague. Fuzzy systems can be used as an integral part of deep learning models by using fuzzy parameters or by using fuzzy logic for selecting training parameters. This novel approach of DNFS has shown possibilities to extract fuzzy rules automatically without the involvement of a human expert solving the "black-box" problem of DNN. Furthermore, DNFS approach shows better accuracy than DNN with the same level of abstraction [26], [27]. Although the concept of DNFS is still very new, but the potential of deep neural network and fuzzy inference model has been implemented in many ways. These powerful architectures are classified into three categories, such as sequential, parallel and cooperative DNFS, as illustrated in Fig. 1. In a sequential architecture (a), the information given to the model flows in a sequence manner using the properties of fuzzy system and deep neural network one by one. While in parallel architecture (b), information is interpreted separately from fuzzy systems and DNN which is later fused at a point to get the final outcome. When it comes to cooperative model (c), the fuzzy systems generate fuzzy values out of crisp values, which goes through the deep neural network for higher level of abstraction and training and an outcome is generated using defuzzification, which again transform fuzzy value into crisp output value.

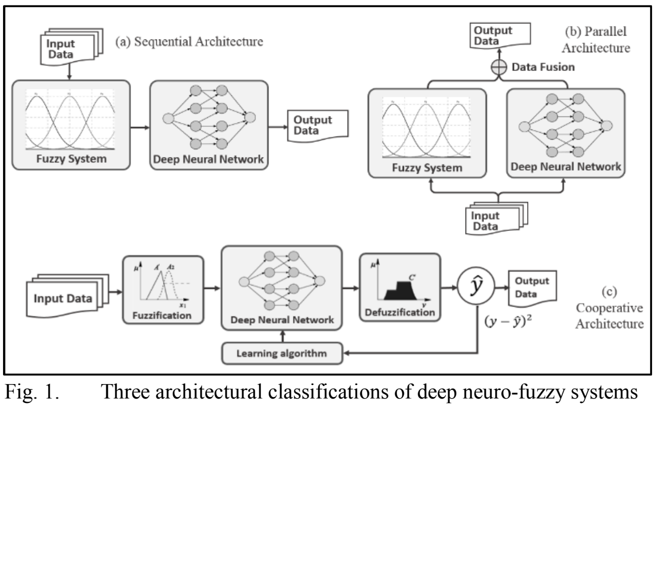

Fig. 1. Three architectural classifications of deep neuro-fuzzy systems

III. Methodology

A careful study of the literature reveals the progressive success of the ANFIS model for solving various real-world classification problems. Yet, this classifier does not perform well with high-dimensional data. This problem was overcome by DNN, which is believed to be a suitable classifier so far for extracting insights from data with a large number of input features. Though DNN outperforms those shallow learning, its application and implementation still have shortcomings due to its "black-box" nature. The classification results made by the DNN model is often criticized for being non-transparent and non-traceable by humans. To address these gaps, some studies are found in literature combining fuzzy inference with DNN as fuzzy deep models to solve various problems.

Therefore, the goal of this study is to propose a Deep Neuro-Fuzzy Classifier (DNFC) combining the idea of ANFIS and DNN into a single architecture in order to solve high-dimensional classification. Hence, this section presents the research methodology of the proposed model.

A flow of research steps that have been taken to shape the methodology is demonstrated in Fig.2, which starts from data collection. In this step, three classification datasets will be taken from UCI (University of California Irvine) Machine Learning Repository [28] and KEEL (Knowledge Extraction based on Evolutionary Learning) [29]. Since the majority of the real-world data are inconsistent and hold missing values, which causes error while analyzing the proposed method. Hence, data preprocessing is an important step to improve the quality of data to achieve maximum accuracy. Once the data is preprocessed, data partitioning has been performed to split the data for training and testing purposes. Therefore, in this study, the 70% of the dataset instances were selected for the training set and the remaining 30% of the dataset instances were chosen for test set due to the reason that the more data that has been used for training the more efficient and reliable results will be produced by the proposed method [30]. Once the data is ready, the next step is to develop the model based on the proposed approach presented in Fig 4. to solve the classification problems. In the end, based on results generated by the proposed model implementing the proposed approach on carefully cleaned data, the performance of the model will be analyzed with the help validation matrix.

Fig. 2. The flow of research steps involved in the methodology of the proposed model

A. *Proposed Approach*

This section discusses the proposed Deep Neuro-Fuzzy Classifier (DNFC) of this study (illustrated in Fig.4) which is divided into three main components: (i) input Membership Function (MF) layer (fuzzification part), (ii) deep neural network (deep learning part), and (iii) output membership function (defuzzification part).

The input layer will take the information from the classification dataset and the network carries the information as follows:

1) Input Membership Function (MF) layer (fuzzification part): In this stage, the fined tuned data after preprocessing has been applied into the model where every input x_i in this layer is an adaptive MF to generate the membership degree of linguistic variables. This stage helps to convert crisp values into the fuzzy membership degree.

$$\mu_{A_i}(x_1),\ \mu_{B_{i-2}}(x_2), \qquad i = 1,2 \text{ or } i = 3,4. \qquad (1)$$

Here, x_1 and x_2 are the two inputs and the MF μ_{Ai} and μ_{Bi} used in this model is Gaussian MFs which holds two parameters (center c and width σ) as in eq. 2.

$$guassian(x;c,\sigma) = e^{-\frac{1}{2}\left(\frac{x-c}{\sigma}\right)^2} \qquad (2)$$

The parameters in this layer are being trained by Gradient Decent (GD) algorithm during the backward pass.

2) Deep neural network (deep learning part): This is the most essential stage in terms of processing the data with a huge number of input features for some high-level abstraction in terms of deep learning. The output coming from the previous stage is used in this layer to initialize the deep neural network. Since the proposed approach transmits the information in two ways, therefore it feed-forward the fuzzy input signals to the hidden layer. The nodes in the hidden layers are interconnected to each other in a way that the information flows in forward pass until the next layer. The input information processed via a single neuron is presented in Fig. 3.

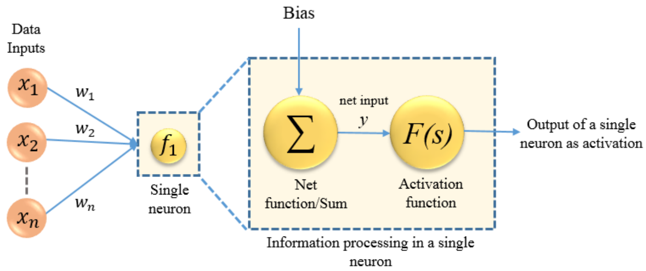

Fig 3. Representation of information processing in a single neuron

The input elements $x_1, x_2, \ldots x_n$ are multiplied with their weight connections w_i, w_2,..., w_n, respectively. The bias is added to calculate the threshold value and a net input y is applied to the activation. Neuron output is associated with input through linear or nonlinear transformation known as the activation function of neuron. The activation function can be linear or sigmoid function. Every neuron uses a weighted sum for a set of inputs and then a nonlinear activation function is used that generates an output value.

3) Output membership function (defuzzification part): From this stage, the deep neural network drives the defuzzification block. The features extracted and learned from the data via DNN will be further processed to defuzzifier generates the output of the proposed model depending on the fuzzy if-then rules exists in the network. A benefit to this system is that these fuzzy rules are used to explain the behavior of the fuzzy system. It is to be noted, however, that expert knowledge is not required, as the network will be set up to automatically extract fuzzy rules from the numerical data. Parameters in this layer are linear parameters that are identified during the training process of the model.

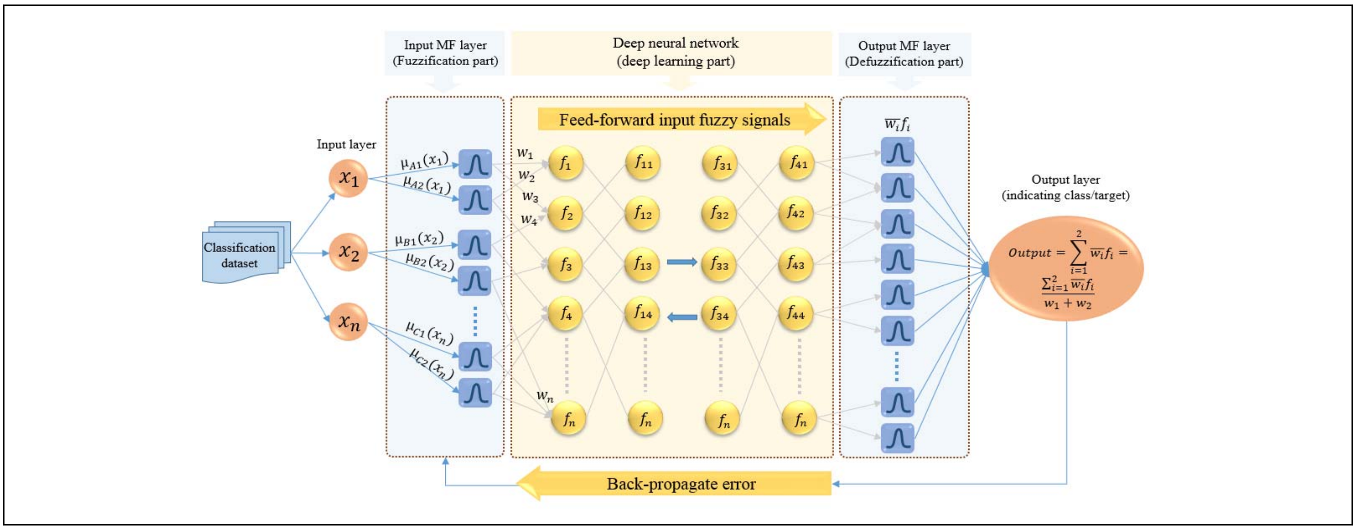

Fig. 4. Proposed Deep Neuro-Fuzzy Classifier (DNFC)

After the process of defuzzification, the next layer is the last layer in which the output of the whole network is calculated as follows:

$$\text{Output} = \sum_{i=1}^{2} \overline{w_i} f_i = \frac{\sum_{i=1}^{2} \overline{w_i} f_i}{w_1 + w_2} \quad (3)$$

This layer simply sums up the outputs of all rules in the previous layer and converts fuzzy values into crisp output. Afterward, the learning algorithm back-propagates the error to update the weights and parameters of the model until the predicted output is similar to the desired output for efficient classification of the data. The weights of the nodes are updated using a gradient-based optimization algorithm.

ALGORITHM 1. BACKPROPAGATION ALGORITHM [31]

1. Backpropagation ($l_e, \alpha, \alpha_{inp}, \alpha_{hid}, \alpha_{out}$)
2. Begin
3. Create a feed-forward network ($\alpha_{inp}, \alpha_{hid}, \alpha_{out}$)
4, Initialize all network weights around 0-0.5 randomly
5. For each $(\vec{x}, \vec{l})$ in l_e,
6. Begin
7. Propagate the input forward through the network:
8. Propagate the errors backward through the network:
9. Calculate $(\delta_k, \delta_\downarrow k \leftarrow o_\downarrow k(1 - o_\downarrow k)(l_\downarrow k - o_\downarrow k))$
10. $\delta_h \leftarrow o_h(1 - o_h) \quad \sum_{k \in outputs} w_{kh}\, \delta_k$
11. Calculated δ_h,
12. Update w_{ji}, $w_{ji} \leftarrow w_{ji} + \Delta w_{ji} \quad and \quad \Delta w_{ji} = \alpha\, \delta_j\, x_{ji}$
13. End
14. End

In a Back-Propagation (BP) algorithm (algorithm 1), x_{ji} is the input and w_{ji} is the weight from unit i to unit j, respectively. Each training example is a pair of the form$(\vec{x}, \vec{l})$. The number of training examples is denoted by l_e. $\vec{x}$ is the input vector, and $\vec{l}$ is the target vector. α defines the learning rate. Whereas the number of units in the input, hidden, and output layers are stated as $\alpha_{inp}, \alpha_{hid}, \alpha_{out}$. The output of each unit u in the network is defined as o_u and the error terms of output unit k and hidden unit h is expressed by δ_k and δ_h, respectively.

IV. EXPERIMENTAL SETUP

The simulation on proposed DNFC was performed on MATLAB toolbox from MathWorks to solve classification problems; thus, three benchmark classification datasets were taken from the UCI and KEEL machine learning repository as red wine quality, statlog and thyroid disease. The details regarding features/attributes, classes and instances of the datasets are presented in the following table (Table I).

TABLE I. OVERVIEW OF CLASSIFICATION DATASETS

Dataset ID#	Dataset Name	Dataset Type	No. of Attributes	No. of Instances
DS1	Red wine quality	Classification	11	1599
DS3	Statlog	Classification	13	270
DS4	Thyroid Disease	Classification	21	7200

Furthermore, the evaluation metrics used in this study to measure the performance of the proposed model are as follows:

- *Mean Square Error (MSE):* Where m is the size of the training dataset, O_{avg} is the average of selected rules' output and O_m^t is the target output of mth training pair.

$$\text{MSE} = \frac{\sum_{i=1}^{m} (O_{avg} - O_m^t)^2}{m} \quad (4)$$

- *Percentage of accuracy:* Where $Correct_N$ and $Instances_T$ are the testing instances correctly identified and a total number of testing instances, respectively.

$$\text{Accuracy Percentage} = \left(\frac{Correct_N}{Instances_T}\right) \times 100 \quad (5)$$

Based on our literature study, we decided on the parameters for our proposed approach to perform the experiments. Since the proposed model is a combination of ANFIS and DNN, the parameter for ANFIS are two MF per input with Gaussian type of MF, output MF is linear, and grid partitioning is used as rule generation method. DNN parameters are set with 0.001 learning rate (α), ReLu activation function, batch size 100 and number of hidden layers are determined during the training process. While the common parameter of the proposed classifier are the maximum number of the epoch as 100, the number of nodes in the output layer is one and we setup error tolerance less than or equal to 0.005 for overall output.

V. Results and Discussions

The experiments were performed on three classification datasets, and the performance of the proposed Deep Neuro-Fuzzy Classifier (DNFC) is evaluated with ANFIS and DNN model based on MSE and percentage of accuracy (Acc%) as in following Table II.

TABLE II. RESULTS OVER ANFIS, DNN AND DNFC FOR CLASSIFICATION DATASETS

Comparison models	ANFIS		DNN		DNFC	
Classification Datasets	**MSE**	**Acc%**	**MSE**	**Acc %**	**MSE**	**Acc %**
DS1	0.0802	95.99	0.0599	97.02	0.0478	97.61
DS2	0.1252	93.74	0.0684	96.58	0.0723	96.38
DS3	0.1509	92.45	0.0417	97.91	0.0401	97.99

According to the results presented in Table II, it can be concluded that the performance of the ANFIS classifier is decreasing as input features are increasing from dataset D1 to D3. While the results generated from DNN and DNFC classifiers are better than ANFIS due to its capability of handling dataset with a large number of input features. However, for D2 the MSE of DNN and DNFC is better than ANFIS, yet it is less than the MSE in D1 even though having more input features. In this situation, the dataset may not be appropriate for deep networks may be one factor. While the results generated for D3 are again showing better accuracy for DNN and DNFC while for ANFIS, the MSE is less than the other two datasets.

Considering the overall generated results from ANFIS, DNN, and DNFC for three classification datasets, it can be concluded that the ANFIS classifier performs well with a small number of input features, whereas DNN performs well when the dataset has a large number of input features plus enough training samples. The most important and noteworthy point of these experiments is that even though our proposed DNFC is showing more or less the same accuracy as DNN, yet it overcomes the "black-box" problem of DNN.

VI. Conclusion and Future Works

In this study, we explained the architecture and working style of ANFIS and DNN and highlighted both advantages and disadvantages of these two commonly utilized classifiers. Since ANFIS is good at solving nonlinear problems, yet the applications of ANFIS are only limited to the problems with less dimensional data. Whereas deep learning methods such as DNN outperforms conventional methods while solving classification problems with a large number of input features due to its higher-level of abstraction and feature abstraction capabilities. On the one hand, the implementation of the DNN classifier upsurged in solving large and complex problems. On the other hand, due to its deep structure performs the optimization of millions of parameters. Therefore, the results generated by DNN started been criticized as non-transparent and not easily understandable by the experts. Recently, researchers have tried to overcome that shortcoming of DNN by using fuzzy logic. Similarly, the goal of this study was also to propose a cooperative structure-based Deep Neuro-Fuzzy Classifier (DNFC) in order to particularly solve classification problems. Based on experimental results using three benchmark classification datasets, we can conclude that the DNN and DNFC classifiers performed well as compare to ANFIS. Though the classification accuracy of DNN and our proposed DNFC is nearly the same, however, it overcomes the problem of non-transparency in DNN.

In this study, we have tried to take the datasets between ten to twenty input features to perform preliminary experiments on the proposed classifier. For future consideration, our next and most important objective is to implement the proposed classifier for solving problems having an even larger number of input features, which is the main focus of developing the proposed classifier. Besides, this study focuses on solving only classification problems. Thus, in the future, we aim to implement the proposed method with problems other than classification problems.

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